

A+: UNDERSTANDING OF NETWORK

TASK: 1

Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. **Hint:** If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

The screenshot shows a web browser window with the address bar displaying '192.168.1.1'. A red box highlights the IP address, and a red arrow points to it with the label 'Router IP Address (Default Gateway)'. The page features a 'Log In with Local Password' form with 'Username' and 'Password' fields. Blue arrows point to these fields with labels 'Enter Router Username' and 'Enter Router Password'. A 'Log In' button and a 'Forgot Password?' link are also visible. At the bottom, a table provides default login credentials, and a note states that these may vary by manufacturer or ISP.

TP-Link Wireless N Router WR840N

Not secure 192.168.1.1

tp-link TP-Link Wireless N Router Model No. WR840N

Router IP Address (Default Gateway)

Log In with Local Password

Username

Password

Log In

[Forgot Password?](#)

Enter Router Username

Enter Router Password

Default Login Credentials (Example)

Username	Password
admin	admin

Note:
Username and password may vary depending on your router manufacturer or ISP.

TASK: 2

Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Entering command prompt:

Writing "ipconfig" the IPv4 address at the bottom last pops up is what our computer's details.

IPv4 Address : 192.168.29.136

Subnet Mask : 255.255.255.0

**Default Gateway : fe80::ae10:7ff:fec2:5bad%23
192.168.29.1**

TASK: 3

Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet

access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

Steps to Set a Static IP

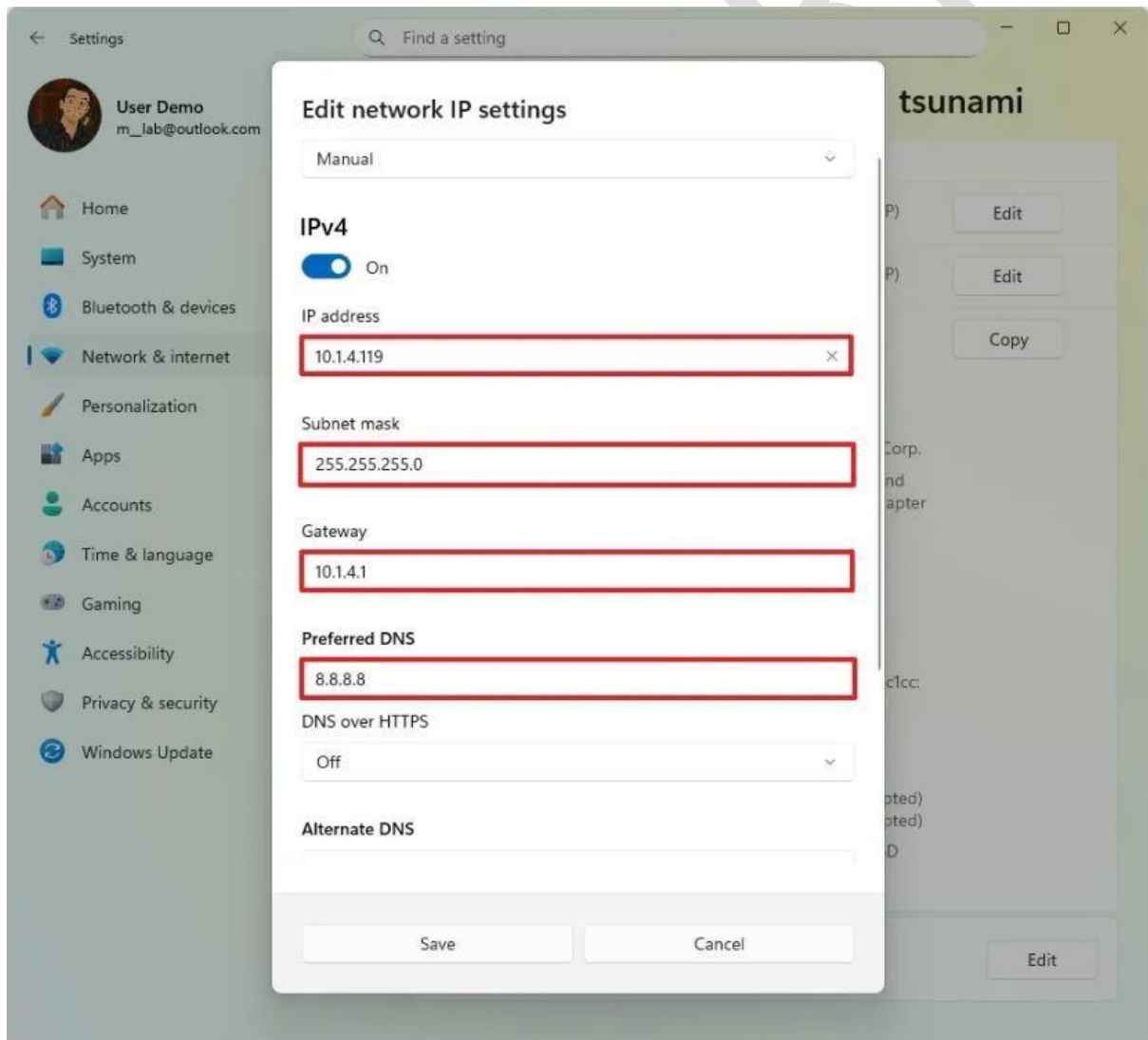
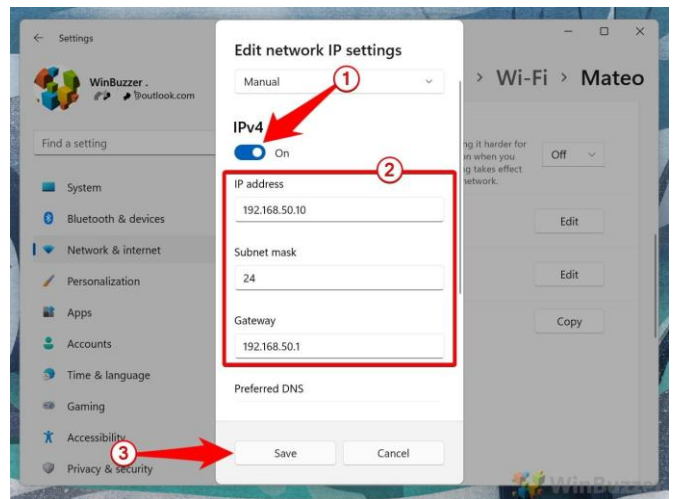
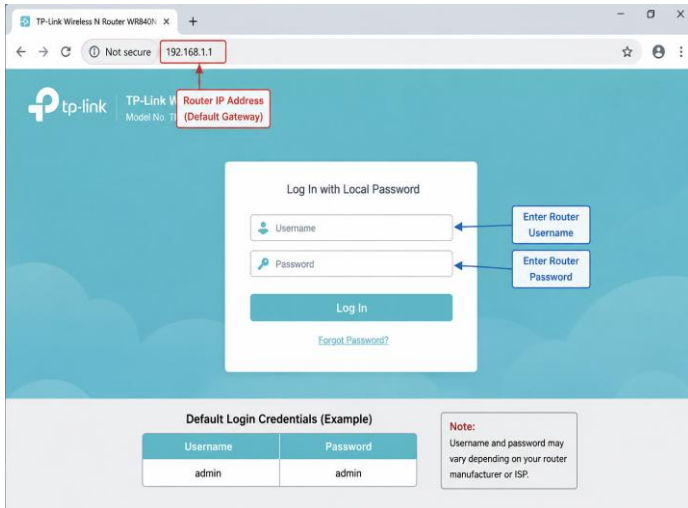
1. Press **Windows + R**, type `ncpa.cpl`, and press **Enter**.
2. Right-click **Wi-Fi** → **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** → **Properties**.
4. Select **Use the following IP address**.
5. Enter:
 - IP address: 192.168.1.50
 - Subnet mask: 255.255.255.0
 - Default gateway: 192.168.1.1
6. Select **Use the following DNS server addresses**:
 - Preferred DNS: 8.8.8.8
 - Alternate DNS: 1.1.1.1
7. Click **OK** and reconnect to Wi-Fi.

Test Internet Access

Open Command Prompt and run:

```
ping 8.8.8.8
```

Then open a website in your browser. If the ping receives replies and websites load, the static IP configuration is working.



Conclusion

Yes, the PC can access the internet using a static IP address. DHCP is not required for internet access; DHCP only automatically provides IP configuration information. After completing the test, the network settings should be changed back to **“Obtain an IP address automatically”** and **“Obtain DNS server address automatically”** to avoid future network conflicts.

NETWORK

TASK: 4

In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Viewing the DHCP Range in a Router:

In the router's admin panel, the DHCP settings can usually be found under:

Network Settings → LAN Settings → DHCP Server

Example DHCP Range

Setting	Example Value
Starting IP Address	192.168.1.100
Ending IP Address	192.168.1.200
DHCP Range	192.168.1.100 – 192.168.1.200

What Happens if Two Devices Get the Same IP Address?

If two devices are assigned the same IP address, an **IP address conflict** occurs. Both devices may experience unstable or lost network and internet connectivity because the network cannot reliably determine which device should receive the data.